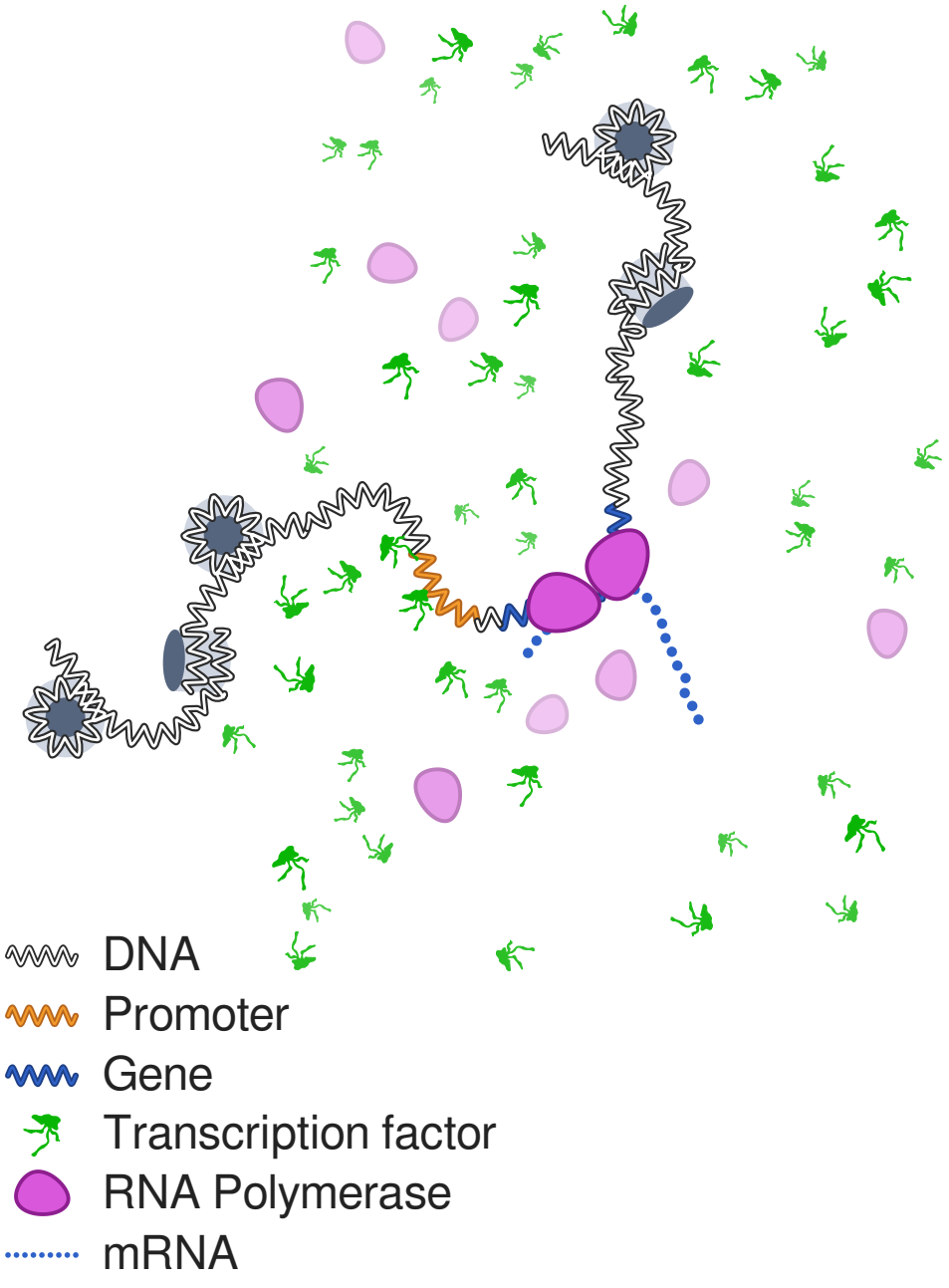
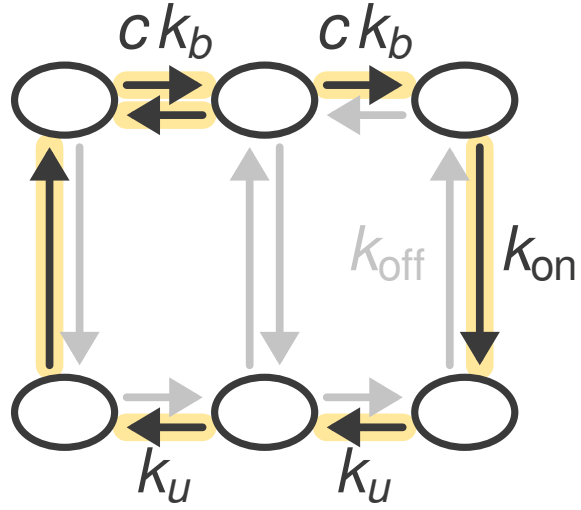
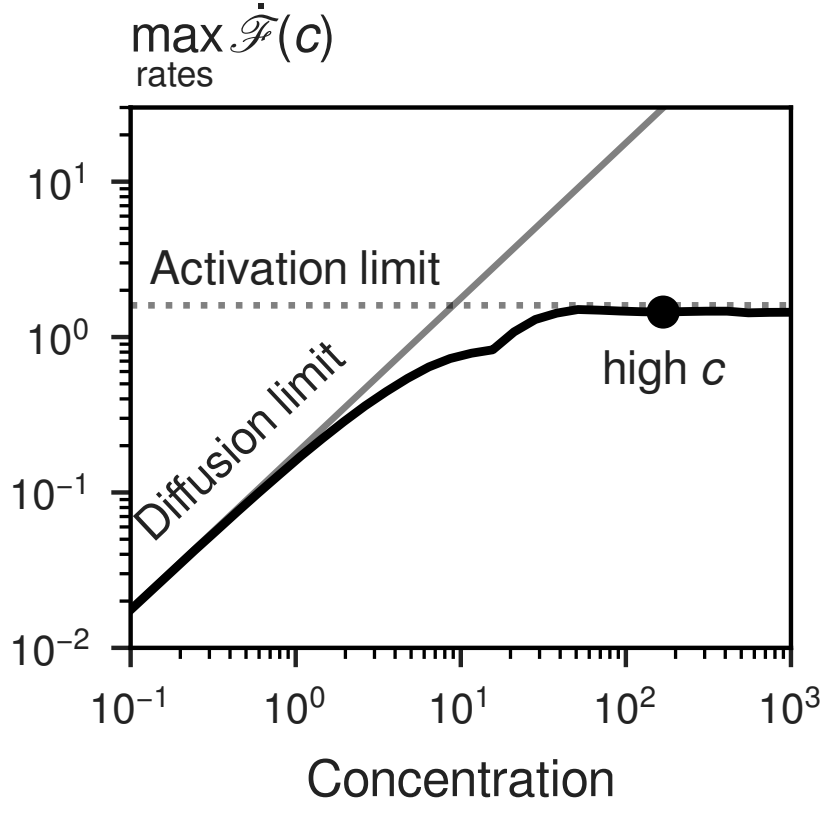

DNA
Promoter
Gene
Transcription factor
RNA Polymerase
mRNA


$c\,k_b$
$c\,k_b$
k_{off}
k_{on}
k_u
k_u
Transcription OFF
Transcription ON


$\max_{\text{rates}} \dot{\mathcal{F}}(c)$
10^{1}
10^{0}
10^{-1}
10^{-2}
Activation limit
Diffusion limit
high c
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Concentration